

Discrete optimization

The curse of dimensionality

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Optimization: principles and algorithms



ÉCOLE POLYTECHNIQUE
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The curse of dimensionality

When we have introduced the transshipment problem, we have seen that some problems can be solved by ignoring the integrality constraints, and the solution would be guaranteed to be integer.

Unfortunately, this property occurs only exceptionally.

There is no optimality condition for discrete optimization

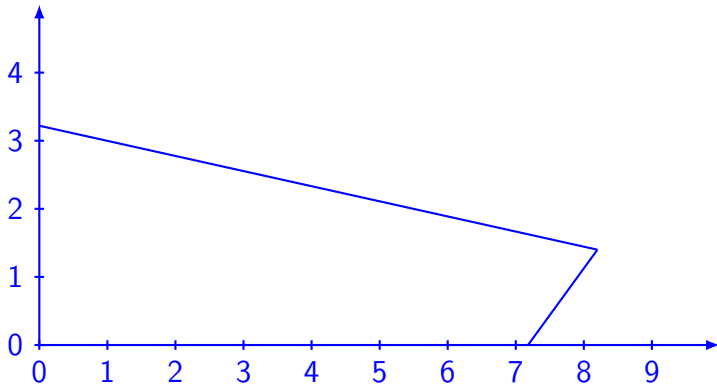
An example

$$\min_{x \in \mathbb{N}^2} -3x_1 - 13x_2$$

subject to

$$2x_1 + 9x_2 \leq 29$$

$$11x_1 - 8x_2 \leq 79.$$



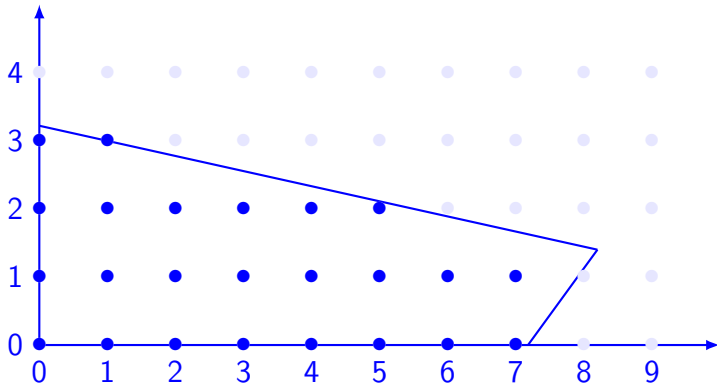
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Enumeration

Solution (1,3) -42

Enumeration

x_1	x_2	$c^T x$	x_1	x_2	$c^T x$	x_1	x_2	$c^T x$
0	0	0	2	0	-6	4	2	-38
0	1	-13	2	1	-19	5	0	-15
0	2	-26	2	2	-32	5	1	-28
0	3	-39	3	0	-9	5	2	-41
1	0	-3	3	1	-22	6	0	-18
1	1	-16	3	2	-35	6	1	-31
1	2	-29	4	0	-12	7	0	-21
1	3	-42	4	1	-25	7	1	-34

Enumeration: the binary knapsack problem

- ▶ n items.
 - ▶ Number of possibilities: 2^n .
 - ▶ For each of them,
 1. check feasibility,
 2. calculate the objective function.
 - ▶ About $2n$ floating point operations.
 - ▶ Processor: 1 Teraflops.
 10^{12}
- ▶ $n = 34$: *1 second*
 - ▶ $n = 40$: *1 minute*
 - ▶ $n = 45$: *1 hour*
 - ▶ $n = 50$: *1 day*
 - ▶ $n = 58$: *1 year*
 - ▶ $n = 69$: *2583 years. Christian Era*
 - ▶ $n = 78$: *1 500 000 years. homo erectus.*
 - ▶ $n = 91$: *10^{10} years. Age of the universe.*

Enumeration: the binary knapsack problem

1 Teraflops

- ▶ $n = 50$: 1 day.
- ▶ $n = 69$: 2,583 years.
- ▶ $n = 78$: 1,500,000 years.
- ▶ $n = 91$: 10^{10} years.

1000 Teraflops

- ▶ $n = 59$: 1 day.
- ▶ $n = 69$: 2.6 years.
- ▶ $n = 78$: 1,500 years.
- ▶ $n = 91$: 10 millions years.

Summary

In the absence of optimality conditions, enumeration is required to identify the optimal solution

The curse of dimensionality is a huge explosion of the running time with the size of the problem.

And waiting for faster computers will not help.

Therefore, explicit and complete enumeration is not a practical algorithm.